

AgriTrace - Installation Guide

AgriTrace Development Team

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1 Introduction

This guide provides step-by-step instructions for setting up the AgriTrace project locally for development and production.

2 Prerequisites

Ensure you have the following installed on your system:

- **Node.js**: Version 18.0 or higher
- **npm**: (Included with Node.js)
- **Git**: For cloning the repository

3 External Services Setup

You will need accounts and API keys for the following services:

3.1 Firebase Setup

1. Create a new project in the Firebase Console.
2. Enable **Authentication** (Email/Password).
3. Enable **Cloud Firestore** in test mode or production (with rules).
4. Enable **Cloud Storage**.
5. Add a **Web App** to your project to obtain the Firebase configuration object.

3.2 Google GenAI (Genkit)

1. Obtain an API key from the Google AI Studio.
2. This is required for the intelligent waste classification features.

3.3 Razorpay Setup

1. Register for a Razorpay account and get your **Key ID** and **Key Secret** from the Settings > API Keys section.

4 Local Installation

4.1 1. Clone the Repository

```
1 git clone https://github.com/harsh-pandhe/AgriTrace.git
2 cd AgriTrace
```

4.2 2. Install Dependencies

```
1 npm install
```

4.3 3. Environment Variables

Create a `.env.local` file in the root directory and populate it based on `.env.local.example`:

```
1 # Firebase Client Config
2 NEXT_PUBLIC_FIREBASE_API_KEY=your_api_key
3 NEXT_PUBLIC_FIREBASE_AUTH_DOMAIN=your_project.firebaseio.com
4 NEXT_PUBLIC_FIREBASE_PROJECT_ID=your_project_id
5 NEXT_PUBLIC_FIREBASE_STORAGE_BUCKET=your_project.appspot.com
6 NEXT_PUBLIC_FIREBASE_MESSAGING_SENDER_ID=your_id
7 NEXT_PUBLIC_FIREBASE_APP_ID=your_app_id
8
9 # Razorpay
10 NEXT_PUBLIC_RAZORPAY_KEY_ID=your_key_id
11 RAZORPAY_KEY_SECRET=your_key_secret
12
13 # Google GenAI
14 GOOGLE_GENAI_API_KEY=your_google_ai_key
```

4.4 4. Running for Development

```
1 npm run dev
```

The application will be available at `http://localhost:3000`.

5 Troubleshooting

- If you encounter issues with `genkit`, ensure you have the `genkit-cli` installed: `npm install -g genkit`
- For Firebase connection issues, double-check your API keys and project ID in `.env.local`.